Personal Expense Analytics

A 12-month data analysis project: cleaning, analysis, insights, forecast and recommendations

Period	January - December 2025
Dataset	617 clean transactions (631 raw rows)
Tools	Python (pandas, matplotlib), Excel (pivot tables)
Data	Synthetic, generated for learning purposes

Headline results

Total income	$60,018
Total expenses	$40,058
Net savings	$19,960
Savings rate	33.3%
Largest category	Rent (44.9% of spending)

1. Objective and Scope

This project answers four questions about one person's finances over a full year: where does the money go, how much is being saved, which categories exceed budget, and what should next month's spending look like.

Deliverables

- Cleaned transaction dataset (CSV) and the raw version it was derived from
- Monthly, category, weekday/weekend and payment-method analysis
- Anomaly flags for unusual months and unusually large purchases
- Four charts, a next-month forecast and three recommendations

Note on data

The dataset is synthetic. It was built to behave like real personal finance data, including seasonal spikes and deliberately messy entries, so the cleaning step is realistic. Replace it with your own bank export to get real findings; the method stays the same.

2. Dataset Description

Column	Meaning	Example
Date	Transaction date	2025-03-14
Description	What was bought or received	Groceries
Category	Spending or income group	Food
Payment Method	How it was paid	Debit Card
Amount	USD, always positive	42.50
Type	Income or Expense	Expense

Sample rows (cleaned)

Date	Description	Category	Payment	Amount	Type
2025-03-09	Ride share	Transport	Debit Card	14.65	Expense
2025-06-20	Groceries	Food	Debit Card	36.48	Expense
2025-07-26	Bus/Metro	Transport	Debit Card	10.41	Expense
2025-08-27	Fuel	Transport	Debit Card	31.32	Expense
2025-09-07	Bus/Metro	Transport	Credit Card	24.52	Expense
2025-10-08	Internet & phone	Bills	Debit Card	95.00	Expense
2025-10-18	Fuel	Transport	Cash	32.26	Expense
2025-11-17	Fuel	Transport	Cash	12.22	Expense

3. Data Cleaning

Raw bank-style data is rarely analysis-ready. Four problems were found and fixed before any numbers were calculated.

Issue	Count	Fix
Exact duplicate rows	14	Removed with drop_duplicates()
Inconsistent category text (case, spaces)	53	Stripped whitespace, converted to Title Case
Mixed date formats (YYYY-MM-DD and DD/MM/YYYY)	25	Parsed each format explicitly, then unified
Missing payment method	9	Filled with "Unknown"

Result: 631 raw rows became 617 clean rows. Dates were parsed with explicit formats because a value like 03/04/2025 is ambiguous and automatic guessing can silently swap day and month.

In Excel, the same steps are: Data > Remove Duplicates; =PROPER(TRIM(C2)) for categories; Text to Columns for dates; and Find & Replace for blanks.

4. Analysis

4.1 Monthly cash flow

Month	Income	Expenses	Savings	Savings rate
Jan	$4,800	$3,425	$1,375	28.7%
Feb	$4,800	$2,956	$1,844	38.4%
Mar	$5,315	$3,308	$2,006	37.8%
Apr	$4,800	$3,331	$1,469	30.6%
May	$4,800	$3,191	$1,609	33.5%
Jun	$5,395	$3,415	$1,980	36.7%
Jul	$4,800	$3,175	$1,625	33.8%
Aug	$4,800	$2,972	$1,828	38.1%
Sep	$5,443	$3,248	$2,195	40.3%
Oct	$4,800	$3,247	$1,553	32.4%
Nov	$4,800	$3,756	$1,044	21.7%
Dec	$5,465	$4,033	$1,432	26.2%
Total	$60,018	$40,058	$19,960	33.3%

Savings rate = (income - expenses) / income. Months with freelance income (Mar, Jun, Sep, Dec) show higher rates.

4.2 Spending by category

Category	Annual total	Share	Monthly average
Rent	$18,000	44.9%	$1,500
Food	$10,666	26.6%	$889
Transport	$3,247	8.1%	$271
Shopping	$3,057	7.6%	$255
Bills	$2,514	6.3%	$210
Entertainment	$1,483	3.7%	$124
Health	$1,091	2.7%	$91

4.3 Budget vs actual (monthly average)

Category	Budget	Actual	Variance	Variance %
Food	$600	$889	+$289	+48.1%
Rent	$1,500	$1,500	$0	+0.0%
Transport	$250	$271	+$21	+8.2%
Bills	$350	$210	$-140	-40.1%
Shopping	$300	$255	$-45	-15.1%
Health	$120	$91	$-29	-24.3%
Entertainment	$200	$124	$-76	-38.2%

4.4 Weekday vs weekend (discretionary spending: all categories except Rent and Bills)

	Days in year	Total	Average per day
Weekday	261	$13,257	$50.79
Weekend	104	$6,286	$60.45

A weekend day carries about 1.2x the discretionary spending of a weekday.

4.5 Payment methods

Method	Total spent	Share
Debit Card	$11,570	28.9%
Credit Card	$11,562	28.9%
UPI/Online	$8,928	22.3%
Cash	$7,610	19.0%
Unknown	$389	1.0%

5. Key Insights

5.1 Top three categories

Rent, Food and Transport account for 79.7% of all spending. Rent alone is 44.9%, and it is fixed, so the real room for change is in the variable categories.

5.2 Spike months

Method: a month is flagged when expenses exceed the mean plus one standard deviation ($3,338 + $304 = $3,642). With only 12 monthly points, a two-SD rule almost never fires, so one SD is used here.

Month	Expenses	Above threshold by
November	$3,756	$114
December	$4,033	$391

5.3 Unusually large purchases

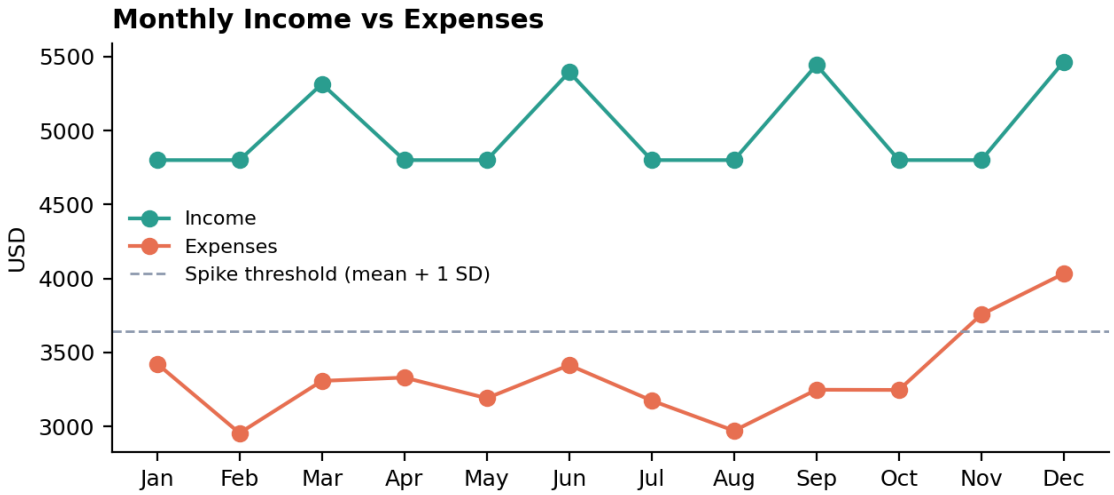
Individual expenses (excluding fixed rent) above mean + 2 SD ($37 + 2 x $39 = $115) are listed below. These are one-off events, not habits.

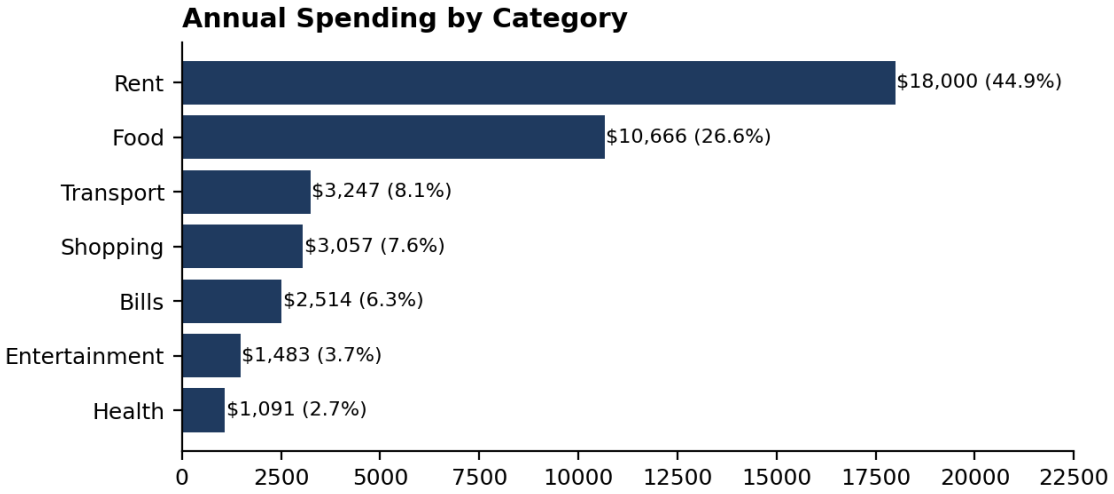
Date	Description	Category	Amount
2025-11-28	Black Friday electronics	Shopping	$480.00
2025-06-17	Dental procedure	Health	$420.00
2025-12-18	Holiday gifts	Shopping	$390.00
2025-12-24	Holiday dinner & travel	Food	$210.00
2025-09-28	Clothing	Shopping	$134.19
2025-04-05	Electricity & water	Bills	$131.67
2025-11-05	Electricity & water	Bills	$129.24
2025-05-05	Electricity & water	Bills	$127.01

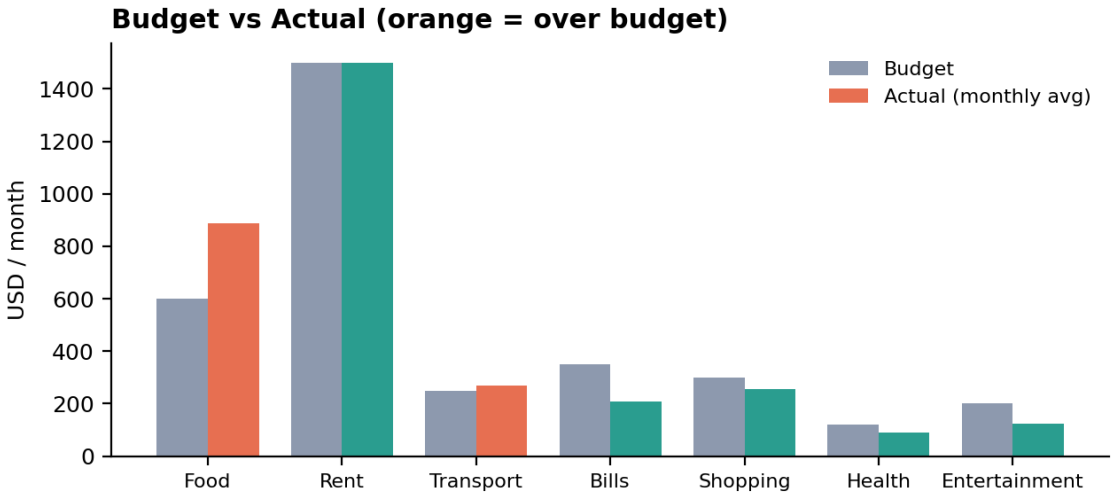
5.4 Budget performance

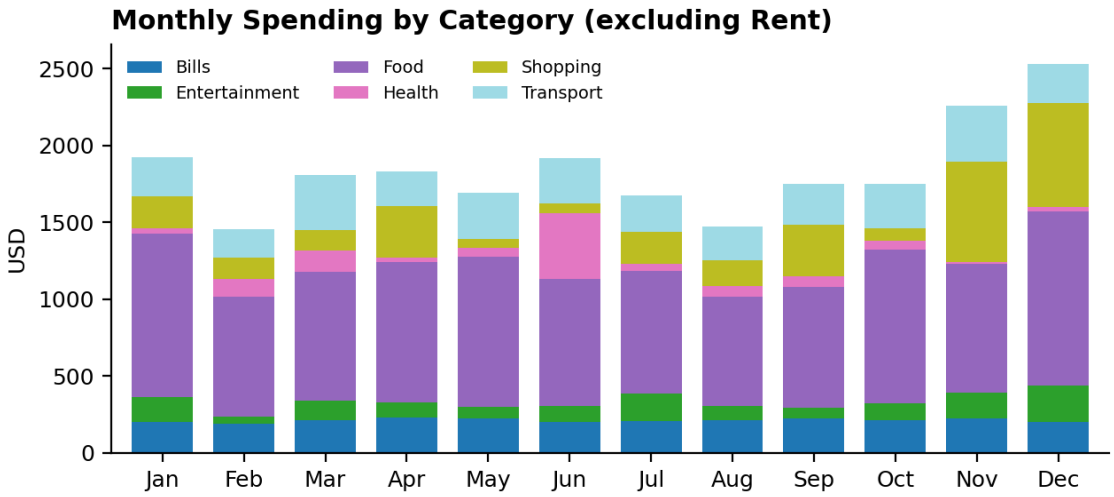
Over budget: Food (+48%), Transport (+8%). Within budget: Rent (+0%), Bills (-40%), Shopping (-15%), Health (-24%), Entertainment (-38%).

6. Visualizations








Monthly Spending by Category (excluding Rent)
Bills
Entertainment
Food
Health
Shopping
Transport
USD
2500
2000
1500
1000
500
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

7. Forecast for January 2026

Two simple methods were compared. The 3-month moving average of expenses is $3,679. A linear trend fitted to all 12 months gives $3,615 (slope +42.5 USD per month). Both are rough guides: with one year of data, seasonality cannot be separated from noise, so treat the result as a planning range of $3,615 to $3,679.

Category view (3-month average, Oct-Dec)

Category	Forecast
Food	$991
Rent	$1,500
Transport	$302
Bills	$212
Shopping	$471
Health	$33
Entertainment	$170
Total	$3,679

The Oct-Dec window includes holiday shopping, so it likely overstates a typical January. If January is a quieter month, expect the lower end of the range.

8. Recommendations

1. Cap the most over-budget category: Food. It runs +48% against plan. Either raise the budget to a realistic figure or set a weekly limit; a 10% cut would save about $1,067 a year.

2. Plan for weekends. Weekend days cost about 1.2x a weekday. A fixed weekend allowance, set on Friday, keeps this visible. Cutting weekend discretionary spending by 10% would save about $629.

3. Build a holiday and health buffer. The spike months came from one-off events (dental work, Black Friday, gifts). Setting aside roughly $198 a month in a sinking fund absorbs these without denting the savings rate.

Target. The current savings rate is 33.3%. Applying recommendations 1 and 2 alone lifts annual savings by about $1,695, roughly 2.8 percentage points.

9. Limitations

- Synthetic data: real behaviour will differ and may include irregular income or refunds.
- Only 12 months, so seasonality and the forecast are approximate.
- Budget figures are assumed targets, not taken from a real plan.
- Category assignment is taken as given; mis-categorized transactions would change the shares.

Appendix: Python script (core analysis)

```
import pandas as pd
df = pd.read_csv("expense_data_raw.csv")
df["Category"] = df["Category"].str.strip().str.title()
df["Date"] = df["Date"].apply(lambda s: pd.to_datetime(
    s, format="%d/%m/%Y" if "/" in s else "%Y-%m-%d"))
df["Payment Method"] = df["Payment Method"].fillna("Unknown")
df = df.drop_duplicates().sort_values("Date")
df["Month"] = df["Date"].dt.to_period("M")

exp = df[df.Type == "Expense"]
inc = df[df.Type == "Income"]

monthly = pd.DataFrame({
    "Income":   inc.groupby("Month").Amount.sum(),
    "Expenses": exp.groupby("Month").Amount.sum()}).fillna(0)
monthly["SavingsRate"] = (monthly.Income - monthly.Expenses) / monthly.Income

by_cat = exp.groupby("Category").Amount.sum().sort_values(ascending=False)
share  = by_cat / by_cat.sum() * 100

thr    = monthly.Expenses.mean() + monthly.Expenses.std()
spikes = monthly[monthly.Expenses > thr]

big = exp[exp.Amount > exp.Amount.mean() + 2 * exp.Amount.std()]

disc = exp[~exp.Category.isin(["Rent", "Bills"])]
disc_weekend = disc.Date.dt.dayofweek >= 5
forecast = monthly.Expenses.tail(3).mean()
```

Excel equivalents

- Pivot table: Rows = Month, Columns = Category, Values = Sum of Amount (filter Type = Expense).
- Savings rate: =(Income - Expenses) / Income
- Spike threshold: =AVERAGE(range) + STDEV.S(range)
- Forecast: =AVERAGE(last three months) or =FORECAST.LINEAR(13, expenses, month_numbers)